Session 5.3

Pretrained Language Models: Encoders That Read, Decoders That Write

Two routes from the Transformer to a measurement: what each was built for, how it works, how its weights are obtained, and what economists measure with it

Based on Chapter 13 of Miao & Feng, *AI for Economists*

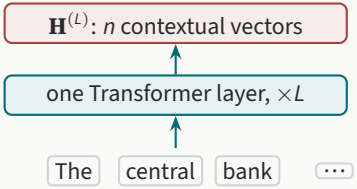
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Where the Pipeline Stands

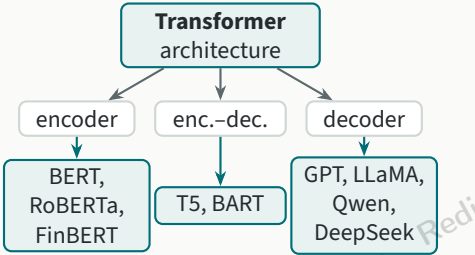


untrained, the attention row for *interest* runs 0.106–0.118: flat

- Deck 5.2 stacked one Transformer layer L times: n tokens in, n contextual vectors $\mathbf{H}^{(L)}$ out. With random weights it reads nothing.
- This deck: two model families, each in four steps: what it was built for, how it reads or writes, how its weights are obtained, what economists measure with it.
- **Pretraining puts language into the weights; how you use them decides the measure.**

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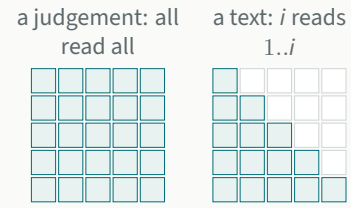
The Family Tree: Architecture, Pretrained Model, Product



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- **Architecture**: Deck 5.2's wiring with random weights. **Pretrained model**: the same wiring, weights fit on a corpus; encoders read, decoders write.
- **Product**: a pretrained decoder taught to follow instructions (**alignment**) and served with default settings.
- **Name the layer you mean: the product is not the architecture.**

Three Jobs, Three Wirings



- The question: what should come back from the model? A judgement about the text, a text, or another sequence.
- A judgement (a label, a score, a vector) → **encoder**: every token reads every other, the square; BERT. A text → **decoder**: a token reads only what is already written, the triangle; GPT.
- Another sequence (a translation; {action: raise, reason: inflation}) → **encoder–decoder**: T5, BART. Measurement tasks fall on the first two.
- **The job fixes the wiring, and the wiring fixes what the model can be asked for.**

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Session 5.3

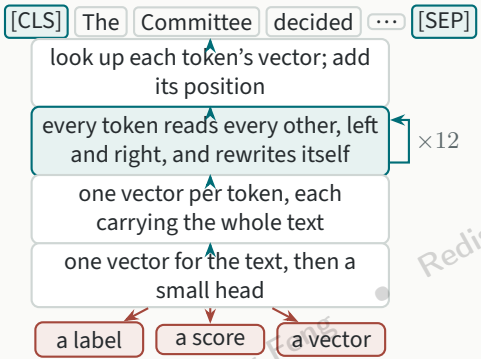
The Encoder Route

Built to read: what BERT is for, how it reads, how its weights are obtained, what it measures



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What an Encoder Is Built For: BERT in Plain Words

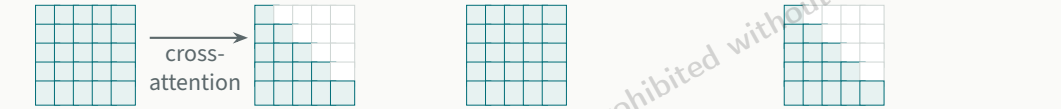


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- BERT reads: text in, numbers out, never more text. A token starts as its word vector plus its position.
- Twelve rounds, no mask: every token reads every other, left and right, and rewrites its vector; at the end each carries the whole text.
- One vector for the text, then a small head: a label, a score, or the vector itself. Weights from guessing hidden words (next frames); the head from a few hundred labels.
- **Text in, numbers out, for a decision made elsewhere: the measurement workhorse.**

How BERT Differs From the Transformer: the Reading Half, Kept Alone

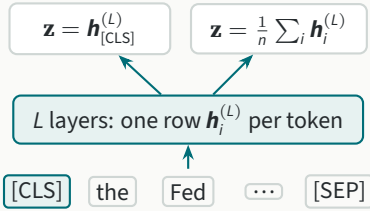


	Transformer, 2017	BERT	GPT, Machine B
reads	square, then triangle	the square: left and right	the triangle: left only
trained to	translate	fill in hidden words	predict the next token
returns	a translation	a judgement	a next-token distribution

- The 2017 Transformer translated: an encoder read the source, a decoder wrote the target, looking back at it (Vaswani et al. 2017).
- BERT keeps the encoder alone, twelve of the same layers, no mask: it reads, never writes (Devlin et al. 2019). GPT keeps the decoder alone: it writes (Radford et al. 2018).
- **Same layer inside; who reads whom, the training task and the output differ.**

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How an Encoder Reads: Both Sides, Then One Vector



cosine of passage vectors	mean	[CLS]
A and B (both on rising rates)	0.72	0.91
A and an unrelated sentence	0.46	0.83
B and an unrelated sentence	0.46	0.85

- Every token reads every other, left and right: the square. A blank needs both sides: *raised*? *cut*?
- Input [CLS] $w_1 \ldots w_n$ [SEP]: [CLS] is a slot with no word whose last row stands for the text; [SEP] marks the end.
- **Pooling**, two ways to one vector: the [CLS] row, or the mean of all rows. Table: a policy statement (A) and an annual-report passage (B), recorded.
- **Untuned, pool by the mean (0.72 related, 0.46 not); raw [CLS] rows are all alike until a task trains them.**

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How the Weights Are Obtained, Stage 1: Fill in the Blanks



both sides needed: *raised*? *cut*?

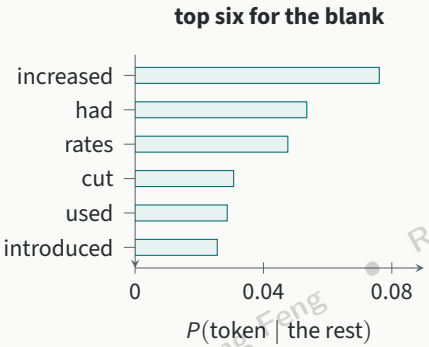
choose 15% of positions; of those 80% → [MASK], 10% → a random token, 10% unchanged

BERT also predicted whether sentence B follows A; RoBERTa found that dropping it "matches or slightly improves" results (Liu et al. 2019).

- Goal: fit the weights on raw text, no labels: hide words and let the text supply the answers (Devlin et al. 2019).
- $\mathcal{M}$: the chosen 15% of positions; $\tilde{w}_{1:n}$: the corrupted input. Maximise $\sum_{i \in \mathcal{M}} \log P_\theta(w_i \mid \tilde{w}_{1:n})$, $P_\theta = \mathrm{softmax}(\mathbf{W}_{\text{out}} \boldsymbol{h}_i^{(L)})$ over the V tokens.
- The token [MASK] never appears in use, hence the 10% kept and the 10% swapped.
- **To fill a blank the model must read both sides, so $\mathbf{H}^{(L)}$ encodes context.**

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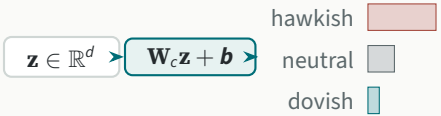
Example: A Pretrained Encoder Fills the Blank



- Multilingual BERT's masked-word head, recorded offline, on "The central bank [MASK] interest rates amid rising inflation.": *increased* 0.076, *cut* 0.031, the true *raised* 0.019; this blank's loss is $-\log 0.019 = 3.97$.
- "Inflation (unemployment) is rising, so the Committee decided to [MASK] the target range": $P(\textit{raise})$ 0.0014 vs 0.0029, the wrong way round.
- **Pretraining delivers the kind of word, not the economics: hence domain text and labels.**

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Stage 2: A Head on Labels

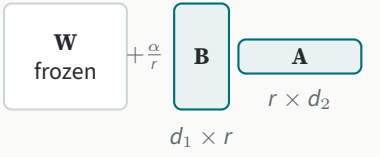


softmax over $C = 3$ classes (bars schematic)

- Goal: a stance for each sentence, learned from N hand-labelled ones.
- Head: $\boldsymbol{p} = \mathrm{softmax}(\mathbf{W}_c\mathbf{z} + \boldsymbol{b})$, $\mathbf{W}_c \in \mathbb{R}^{C \times d}$; loss $-\frac{1}{N}\sum_i \log p_{i,y_i}$: a multinomial logit on the encoder's features.
- Counts: the head has $C(d+1) = 2,307$ numbers ($d = 768$); the encoder 177,853,440. Full fine-tuning moves the encoder too.
- **A few hundred labels can pin down the head, not the encoder.**

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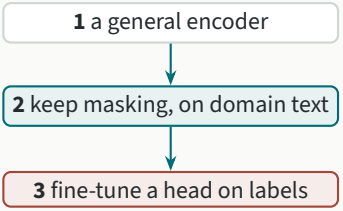
Three Ways to Fine-Tune



- **Full**: update all 177,853,440 weights. Closest fit; with few labels it overfits and loses the general representation.
- **Frozen encoder**: train the head alone. Cheap; the representation cannot adapt.
- **Low-rank adaptation** (LoRA; Hu et al. 2022): $\mathbf{W}' = \mathbf{W} + \frac{\alpha}{r}\mathbf{BA}$, $\mathbf{A}$ random, $\mathbf{B} = \mathbf{0}$ at the start, so training begins at the pretrained model. At $d_1 = d_2 = 768$, $r = 8$: 12,288 numbers against 589,824, 2.1%.
- **The change a task needs has low rank: one base model, one small versioned adapter per task.**

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Domain-Adaptive Pretraining: Same Objective, Different Corpus



- Continue the masking objective on domain text before adding a head (Gururangan et al. 2020): *hawkish* is a stance on war in general text, on rates in central-bank text.
- FinBERT (Araci 2019): BERT, then masking on TRC2-financial (46,143 Reuters articles, over 29M words), then a three-way head on the Financial PhraseBank (4,845 sentences).
- **Which corpus the model read is part of what the measure means.**

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Encoders in Economic Research

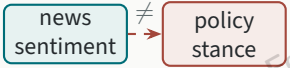
read text, return …	model	what the number means
news sentiment, per sentence	FinBERT (Araci 2019): a three-way head on the Financial PhraseBank	positive, negative or neutral financial news
agent and sign, per central-bank sentence	CentralBankRoBERTa (Pfeifer & Marohl 2023): two classifiers	which of five agents a sentence concerns, and whether it is good for them
hawkish–dovish stance	a head fine-tuned on hand-coded sentences (Deck 5.4)	the coders' definition of stance, carried to every sentence
the class of a 10-K risk paragraph	a head with classes such as regulatory, market, operational	the class list the researcher wrote
a vector, for similarity	PaECTER, an encoder trained on patents (Ghosh et al. 2024); Session 6's index	the cosine of pooled vectors: nearness, not sameness

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- **Every row is one head or one cosine on $\mathbf{H}^{(L)}$, the same number at every run; what it means is the training labels and the corpus.**

Example: A Domain Model's Labels Are Its Definition

sentence, written for this slide	positive	negative	neutral
The Committee decided to raise the target range for the federal funds rate.	0.12	0.03	**0.86**
The Committee decided to lower the target range for the federal funds rate.	0.05	**0.50**	0.45
The Committee decided to maintain the target range for the federal funds rate.	0.06	0.03	**0.91**
The labor market has remained strong, and job gains have been solid.	**0.95**	0.03	0.02
Economic growth has slowed, and business investment has weakened.	0.01	**0.97**	0.02



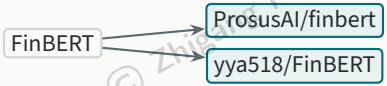
- `ProsusAI/finbert`, recorded offline. A hike reads neutral (0.86), a cut negative (0.50): news for markets, not policy stance.
- **A classifier measures what its training labels meant: read the label set first.**

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Two FinBERTs, and a Model With Two Classifiers

identifier	its paper	what it is
`ProsusAI/finbert`	Araci (2019)	financial news, then a three-way head on the Financial PhraseBank
`yya518/FinBERT`	Yang, UY & Huang (2020)	a different model of the same name, on financial communications
CentralBankRoBERTa	Pfeifer & Marohl (2023)	*two* classifiers: five macroeconomic agents, and a sign for each

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one name, two models

- **Load a model by its identifier, cite the paper with it, and read its label set.**

Session 5.3

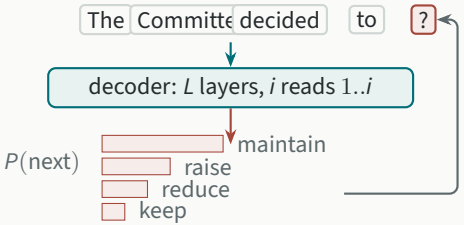
The Decoder Route

Built to write: what GPT is for, how it writes, how its weights are obtained, what it measures



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What a Decoder Is Built For

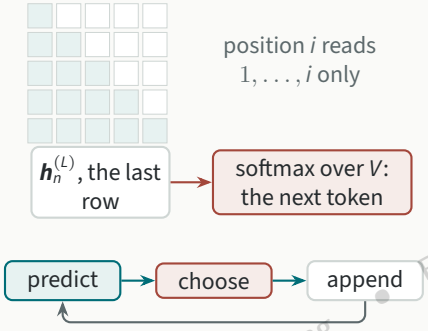


schematic. Choose one, append it, ask again

- Built for writing: given a prefix, a probability for every vocabulary token as the next one (GPT, Radford et al. 2018).
- With scale the purpose grew (Radford et al. 2019; Brown et al. 2020): write a task as text to be continued, “Stance:”, and one model serves many tasks with no head to fit.
- In: a prefix. Out: one distribution. Choose a token, append it, repeat: a text.
- **A decoder turns a prompt into text; measuring with it means asking in the right form.**

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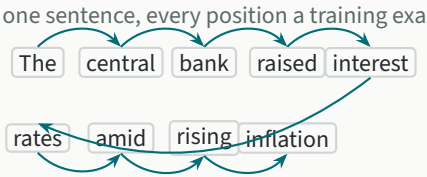
How a Decoder Writes: One Row, One Distribution, One Loop



- The encoder's blocks with the causal mask: each row sees only its past. The last row, through the output layer: $P_\theta(w_{n+1} \mid w_{1:n}) = \mathrm{softmax}(\mathbf{W}_{\text{out}} \boldsymbol{h}_n^{(L)})$.
- Generation loops: predict, choose one token, append it, predict again. **Greedy**: the top token.
- **Temperature** τ divides the scores s_w before the softmax: $\tau \to 0$ is greedy, $\tau > 1$ flattens. Top-k, top-p: draw from the best k, or the smallest set with mass p.
- **The model returns a distribution; a decoding rule turns it into text.**

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How the Weights Are Obtained: Predict the Next Token



8 predictions from nine tokens:
The→central, …, rising→inflation.
Masking 15% of the same sentence gives about 1.35.

- Objective: maximise $\sum_i \log P_\theta(w_i \mid w_{<i})$ over the corpus: at every position, the probability of the token that actually came next (**causal language modelling**).
- No labels: the text supplies its own answers, as the blanks did for the encoder; here every position is an example, not 15% of them.
- The price: only the left context, so position i never reads what follows it.
- **A denser training signal and one-sided reading: that trade separates GPT from BERT.**

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Example: A Next-Token Distribution, Recorded

next character	P	score
`m`	0.532	6.34
`r`	0.084	4.49
`k`	0.072	4.35
`t`	0.061	4.18
`i`	0.049	3.95
`a`	0.038	3.69

The Committee decided to␣ m ?

- **Machine B**: Session 1's text generator, trained on the 245 FOMC statements; 3 layers, $d = 128$, 631,424 parameters, 78 characters as tokens.
- After "The Committee decided to" it puts 0.53 on `m` (maintain), then `r` (raise, reduce) and `k` (keep). The top character each time: "… to maintain the target range for the federal funds rate at 0 to".
- **A fitted decoder is this table at every step; the score column is what temperature rescales. Lab C4 runs it.**

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Example: Decoding Rules on Machine B's Scores

next	$\tau = 0.5$	$\tau = 1$	$\tau = 2$
`m`	0.93	0.53	0.19
`r`	0.02	0.08	0.07
`k`	0.02	0.07	0.07
`t`	0.01	0.06	0.06

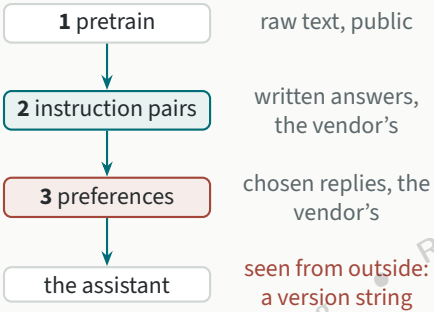
$p_\tau(w) \propto \exp(s_w/\tau)$, the last frame's

$\tau \to 0$ greedy | $\tau = 1$ the model | $\tau > 1$ flatter

- Columns: the same scores divided by τ before the softmax. At $\tau = 0.5$ the top character takes almost everything; at $\tau = 2$ the mass spreads.
- Sampled at $\tau = 2$, Machine B writes "neiqvelopments, both polifle advi2$105. Biral 2 1.S5 fillp4": the spelling of the corpus, the sense gone.
- **For measurement set $\tau = 0$, and report τ, the seed, the version and an input hash.**

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Stage 2 in One Frame: From Continuation to Answers



- A pretrained decoder continues text: asked a question, it may go on with another question (illustrative). Fluent, not an answer.
- Stage 2: the next-token objective again, on instruction–answer pairs people wrote. It fixes form: format, and when to decline.
- Stage 3: people see two replies and pick one; the model moves toward the preferred kind, kept close to stage 2 (Ouyang et al. 2022).
- **Both data sets are private and rolling: a chat product is a moving instrument, and the version string is part of the measure.**

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Decoders in Economic Research

read text, write …	example	what you must report
a label, zero- or few-shot	Hansen & Kazinnik (2023): GPT-4 labels sentences of the 2010–2020 FOMC statements hawkish, neutral or dovish; matched human coders better than a fine-tuned encoder; moved with prompt wording and version	the prompt, the version, agreement with a hand-coded sample (Deck 5.4)
fields, from raw text	dates, parties, amounts from prospectuses, court rulings or job postings, into a panel	the schema, and a checked subsample
a score for a construct	policy uncertainty or financing pressure, one number per document from a rubric	the rubric, $\tau = 0$, several raters (Decks 5.4–5.5)
answers in a persona	a model filling in a survey as a household or a firm	Deck 5.5

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- **Every row is a continuation of a prompt, no parameter changed; so the measure is the prompt, the decoding rule and the version.**

In-Context Learning: The Task Goes in the Prompt

Label the monetary-policy stance of the Federal Open Market Committee statement below. Answer with exactly one word: HAWKISH if it leans toward tighter policy, DOVISH if it leans toward easier policy, NEUTRAL if it leans neither way. Output the single word and nothing else.

STATEMENT: [the statement of 1995-02-01: "The Federal Reserve Board today approved an increase in the discount rate from 4 3/4 percent to 5 1/4 percent, effective immediately." …]

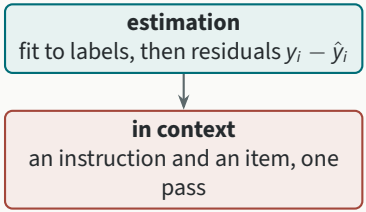
continuation → HAWKISH

recorded: claude-sonnet-5, 2.1.278 (Claude Code), 2026-09-20

- **Zero-shot**: an instruction and the item, as here. **Few-shot**: add worked examples first, e.g. "… decided to raise the target range. HAWKISH" (written for this slide).
- No parameter changes: the label is the model's continuation of the prompt.
- **The prompt is the estimator's specification: freeze it and publish it.**

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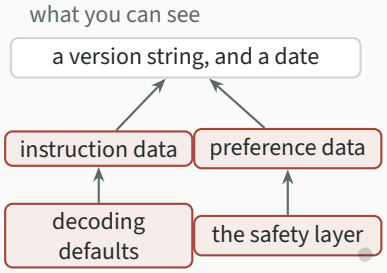
In-Context Labels Carry No Internal Error Measure



- Estimation fits parameters to labelled data and leaves regression residuals $y_i - \hat{y}_i$ to inspect.
- In-context labelling fits nothing: the model returns HAWKISH and no quantity says how likely it is to be wrong.
- Machine B's fluent statements had the same property: no target to subtract them from.
- **So the error is measured outside, on a hand-coded sample. Deck 5.4 builds it.**

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The Version Is the Instrument



- Instruction and preference data are the vendor's, private and rolling; from outside you see a version string, `gpt-4-0613` or `gpt-4o-2024-08-06`.
- Two studies that both "used GPT-4" may have used different instruments, and neither can tell. Deck 5.5 scores one frozen rubric with several raters.
- **An aligned model used for scoring is a time-varying instrument: the version is part of the measure's definition.**

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Session 5.3

Measuring With Either Family

The comparison, the limits that stay, and the lab



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Two Paragraphs, Two Families

A, a policy statement: To counter rising inflation pressure, the central bank decided to raise the policy rate by 25 basis points. The Committee judges that maintaining a restrictive stance is appropriate until inflation returns to its target.

B, an annual report: During the reporting period, the company's financing costs increased as market interest rates rose. Management expects tighter credit conditions to put some pressure on next year's capital expenditure.

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- **BERT**: pool each into $\mathbf{z}_A, \mathbf{z}_B$: cosine 0.72, 0.46 against unrelated text.
- **GPT** (illustrative): "… decided to" continues *raise*; asked A's stance, "hawkish"; asked B's risk type, {risk type: rising financing costs}.
- **BERT returns the same number every time; GPT moves with the prompt, decoding and version.**

Five Limits, Five Architectural Roots

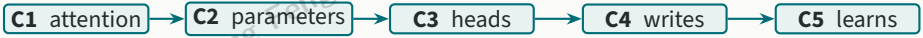
limit	root in the architecture	what it does to a design
the cutoff	knowledge sits in the weights, fixed on a date	put the document in the prompt
hallucination	the objective rewards form, not truth	check every citation and number
the window	attention costs $O(n^2)$ (Deck 5.2)	blocks sever cross-references
lost in the middle	attention favours the ends	position is part of the error
prompt sensitivity	the prompt is the prior	freeze it, and vary the order

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- None is a defect awaiting a patch; each follows from how the machine is built (the middle: Liu et al. 2024; the order: Brucks & Toubia 2025).
- **Deck 5.4 turns them into a protocol; Session 6 repairs four by retrieval.**

Lab, Parts C4–C5: Machine B Writes, Then Learns From Nothing

cell	what you run	the question you answer
C4	Machine B writes 200 characters after “The Committee decided”, at three temperatures	Which dial moves the house phrasing, and which the spelling? What did the objective never ask for?
C5	a fresh model of the same architecture, sixty training steps on the same corpus, held out by whole statement	Where does the loss start ($\ln 78$ nats per character), and what has it learned after sixty steps?



- **C4 reproduces the next-character and temperature tables; C5 starts where every model starts, at the uniform distribution.**

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